Predictive Analysis of Obesity Status

Qiyang Chen, Xinyao Chen, Riya Hariyaplar, Yutian Yang, Zijia Zhang

Statistics 101C Final Project

December 16, 2024

Abstract

Obesity represents a significant public health concern in the United States. This project investigates the related factors of obesity status through statistical analysis. Our study employs advanced statistical learning models to predict obesity status using a comprehensive numerical and categorical dataset. This report presents a systematic and detailed methodology of our classification model development process.

The final model utilizes random forest. By the time the Kaggle competition ended, we achieved a Kaggle competition score of 98.2%, with our team ranking 35. After the presentation, we learnt from other groups and further improved our model and finally concluded with a final random forest model with 200 trees and 7 predictors per split achieved a testing accuracy of 98.3%.

Introduction

Obesity has emerged as a critical public health challenge in the United States (Jeffery & Utter, 2003; Luckhaupt et al., 2014; Restrepo, 2022), with far-reaching implications for individual health and societal well-being (Chatterjee et al., 2020). This condition is not merely a result of individual choices but is influenced by a complex interplay of environmental, occupational, and socioeconomic factors (Jeffery & Utter, 2003; Joshu et al., 2008; Luckhaupt et al., 2014). Recent studies have shown that obesity is closely linked to various health conditions, including cardiovascular diseases, chronic obstructive pulmonary disease, type II diabetes, hypertension, and depression (Chatterjee et al., 2020).

The prevalence of obesity has been exacerbated by modern workplace conditions, with studies indicating that factors such as long work hours, hostile work environments, and specific

occupational sectors contribute significantly to obesity rates (Luckhaupt et al., 2014). The COVID-19 pandemic has further complicated this issue, leading to increased body mass index (BMI) and obesity rates among U.S. adults, accompanied by changes in lifestyle behaviors such as exercise patterns, sleep duration, and alcohol consumption (Restrepo, 2022).

Socioeconomic factors play a crucial role in obesity distribution, with different patterns observed between developed and developing nations. In developed countries like the United States, obesity tends to affect economically vulnerable groups more severely, with women in lower socioeconomic groups experiencing nearly double the impact compared to men (Chatterjee et al., 2020). The relationship between obesity and mental health, particularly depression, has also been established, though the directionality of this relationship remains complex and subject to ongoing research (Blasco et al., 2020).

Understanding and addressing obesity requires a comprehensive approach considering these multiple contributing factors. Machine learning algorithms have emerged as powerful tools for analyzing and predicting obesity-related risks and consequences, offering new insights into this multifaceted public health challenge (Chatterjee et al., 2020).

Data Exploration and Preprocessing

Our initial data analysis focused on understanding the dataset structure and preparing it for modeling. We comprehensively examined missing values, variable distributions, and correlations between features.

1. Dataset overview

The dataset consists of 29 predictors with 32,014 observations in the training set and 10,672 observations in the testing set. The maximum error rate is the proportion of minority cases in our training data, which is around 39%. The dataset contains two types of variables:

Numerical Variables (11 total):

- Health metrics: Age, Height, Resting Blood Pressure, Cholesterol Level, Maximum Heart Rate, Average Glucose Level
- Lifestyle factors: Frequency of vegetable consumption, Number of main meals, Daily water intake, Physical activity frequency, Technology use time

Categorical Variables (18 total):

- Demographics: Gender (Female/Male), Race (Asian/Black/Hispanic/White), Marital status (Yes/No), Residence type (Rural/Urban)
- Health conditions: Heart Disease (Yes/No), Hypertension (Yes/No), Stroke (Yes/No)
- Lifestyle factors: High-calorie food consumption (Yes/No), Between-meal eating (No/Sometimes/Frequently/Always), Smoking status (Yes/No), Sweet drink consumption (Yes/No), Caloric intake (No/Sometimes/Frequently/Always), Transportation mode (Automobile/Bike/Motorbike/Public Transportation/Walking)
- Medical indicators: Fasting Blood Sugar (Yes/No), Resting ECG (LVH/Normal/ST), Exercise status (Yes/No)

- Other: Family history of overweight (Yes/No), Work type (Children/Government job/Never worked/Private/Self-employed)

2. Data Cleaning

Firstly, we used the aggr() function to analyze missing values and their patterns across both datasets. The training set showed a consistent proportion of missing values around 8%, ranging from 7.65% for stroke status to 8.33% for work type. The testing set showed similar patterns, with physical activity frequency (FAF) at 7.4% and work type at 8.6%. Since all predictors had identical rates of missing values, we couldn't justify removing any single predictor based on missing data alone. This led us to explore data-cleaning strategies and we came up with two methods.

Method 1: Complete case analysis using na.omit()

Using na.omit() would reduce the training dataset to 2,933 observations and the testing dataset to 986 observations. Given this substantial reduction in valuable data points, we decided this approach was not optimal.

Method 2: Multiple imputation using mice()

We applied mice() once, successfully imputing all missing values in training and testing datasets. Therefore, subsequent analysis and modeling are all based on this imputed dataset.

3. Exploratory Data Analysis

❖ Density plots for numerical variables

The density plots of all predictors, colored by the two levels of obesity status, show significant or complete overlap with each other. This indicates that individual predictors show little to no differentiation between obesity status levels. Below are the density plots for the four most promising numerical predictors (Figure 1) with the least overlaps between 2 levels.

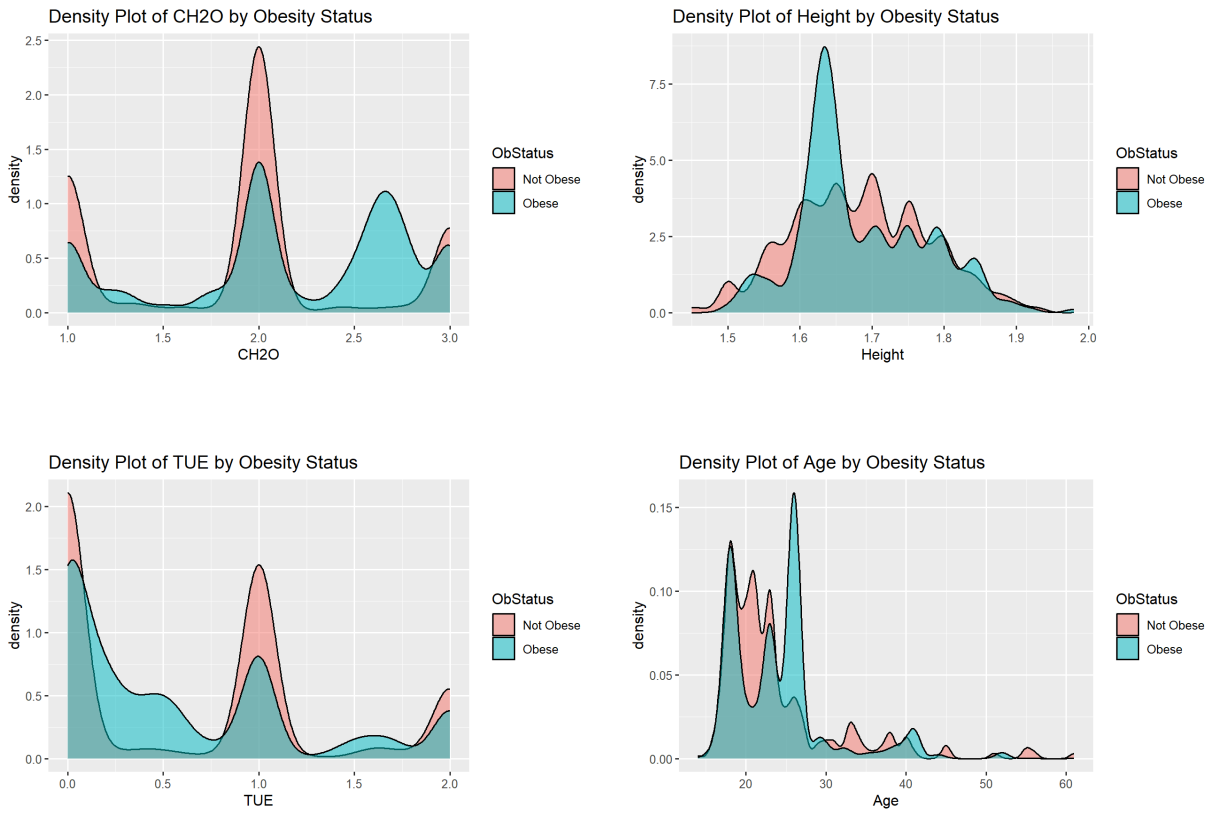

Figure 1. Density plots for 4 most promising numerical predictors

❖ Stacked bar charts for categorical variables

Analysis of categorical variables shows more promising results. Eight categorical predictors demonstrate strong relationships with obesity status: family history of overweight, caloric intake, frequency of vegetable consumption, stroke status, method of transportation, consumption of food between meals, sweet drink consumption, and work type. Four of their charts are shown below (Figure 2). Surprisingly, smoking status shows no significant difference in obesity status.

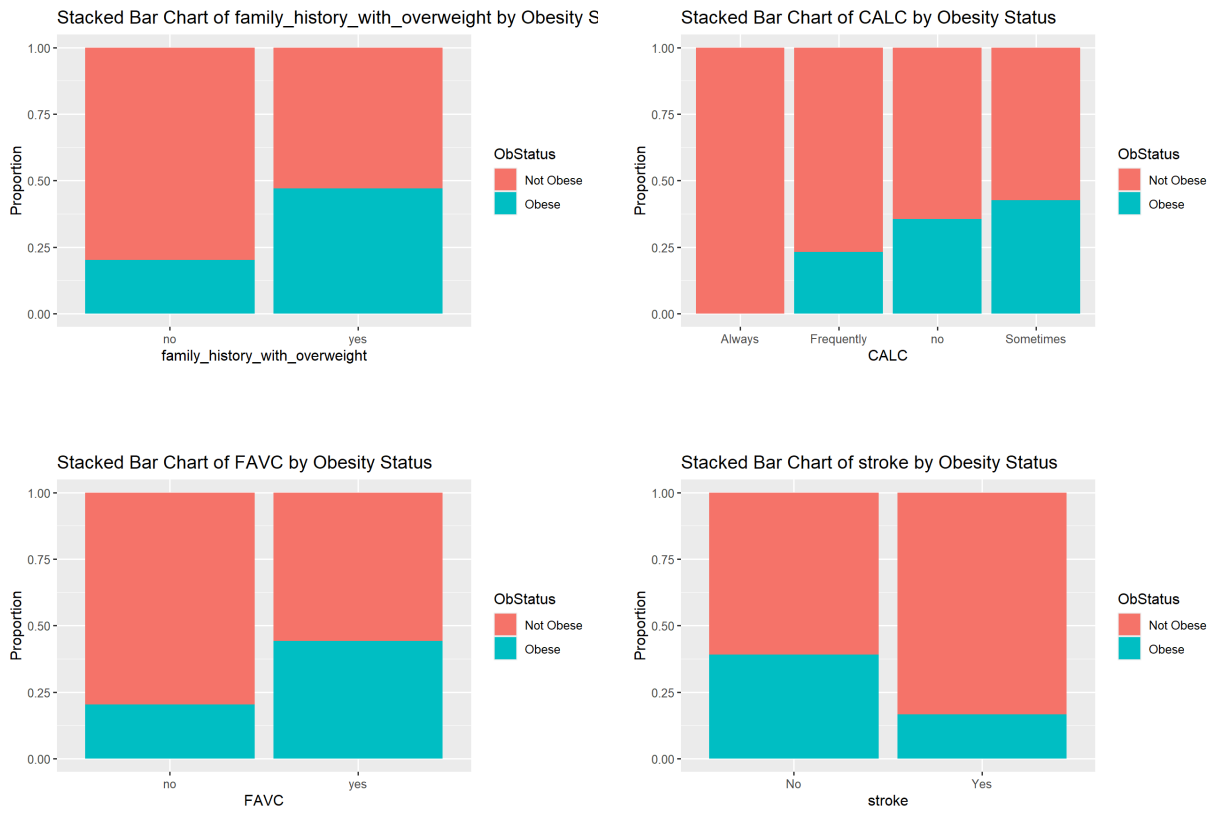

Figure 2. Example Stacked Bar Charts

❖ Multicollinearity analysis

We then assessed the presence of multicollinearity among the predictors by calculating the Variance Inflation Factor (VIF) values. VIF is a key indicator for detecting multicollinearity, where higher values suggest strong linear relationships among predictors. The VIF results, as shown in Figure 3, indicate that all variables have VIF values roughly less than or equal to 2. This suggests that there is no significant multicollinearity issue among the predictors.

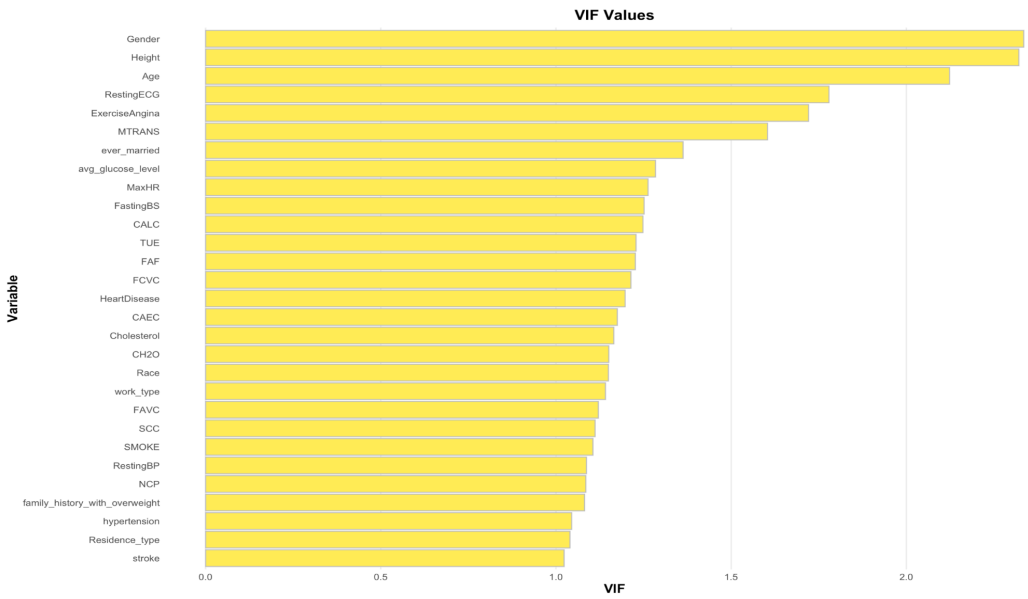

Figure 3. VIF plot for all predictors

Based on this preliminary analysis, we conclude that multicollinearity is not a major concern. However, we will further refine our feature selection during the modeling process to ensure model efficiency and avoid overfitting.

Methods & Models

We applied a comprehensive set of methods using R with a seed of 500 to analyze the data and evaluate their predictive performance. The models were primarily grouped into three categories:

1. **Probabilistic models**

Probabilistic models are based on probabilistic assumptions and leverage distance or probability distributions for prediction. Specifically, we implemented: the **Naive Bayes classifier** and the **K-Nearest Neighbors (KNN) classifier.**

The NB classifier predicts outcomes based on Bayes' theorem. Our implementation of Naive Bayes achieved a testing accuracy of 61.61%. **KNN** is a non-parametric, distance-based method that classifies data points based on their closest neighbors. By combining a k=10 cross-validation with KNN modeling, we achieved a testing accuracy of 72.27%. The Naive Bayes and KNN classifiers showed poor performance, achieving accuracies only marginally above the maximum error rate of 39%. These results indicate that both models struggled to identify meaningful patterns, leading us to explore alternative approaches.

2. **Linear models**

We begin with a full **logistic regression** model that includes all predictors, then improve it using **stepwise selection** (bidirectional) with both AIC and BIC criteria. We also implement regularization methods, **Ridge and Lasso regression**, using cross-validation to determine the optimal penalty term lambda. For dimension reduction, we apply **Linear Discriminant Analysis (LDA)** and **Quadratic Discriminant Analysis (QDA)**. While QDA is not strictly a linear

model, we include it here due to its close relationship with LDA.

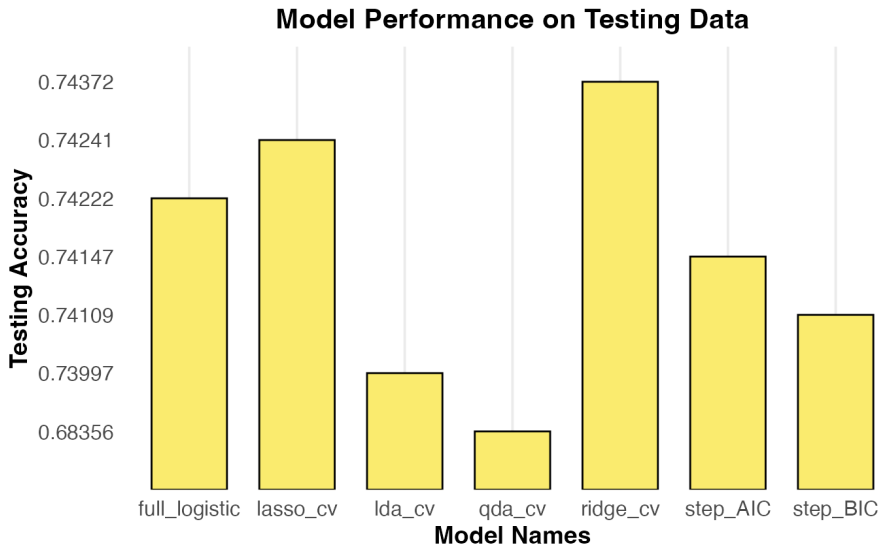

Figure 4. The summary graph for testing accuracies of the models

Test accuracy results are displayed in Figure 4 above. While the regularization method achieved the highest accuracy among linear models, all models performed similarly, with accuracies ranging from 68% to 75%. Given these modest results, we explored non-linear models, for example, splines and random forests, to potentially achieve better accuracy.

3. Splines

Spline regression uses piecewise polynomial functions to capture non-linear relationships in the data, making it highly flexible for complex datasets. Our spline model achieved accuracy of 98.08%, indicating its ability to fit data exceptionally well in cases with strong non-linear patterns. The primary advantage of splines is their adaptability, which ensures smooth transitions between segments. However, the method may overfit if not properly regularized, and it requires careful selection of the number. Despite these challenges, its superior accuracy in this case demonstrates its power for modeling intricate relationships.

4. Trees

Our decision tree model uses four key predictors, daily water consumption (CH2O), physical activity frequency (FAF), Age, and Height, with 7 terminal nodes determined through cross-validation. Although the model achieved a testing accuracy of 78.4%, this single-tree approach had limitations. These limitations led us to explore more sophisticated ensemble methods, bagging and random forests, which use multiple trees to overcome the constraints of individual decision trees.

5. Random Forests

The random forest model is compelling in terms of studying the dataset. It evaluates the data from different perspectives every time it grows a tree. On the other hand, since it incorporates bootstrapping in the process of training the model, it largely prevents overfitting and performs decently on unseen data. We implemented a bagging model with all predictors and 500 trees, achieving 97% accuracy. While bagging improves upon single trees by averaging predictions, random forests typically provide better results with more efficient models. Using cross-validation, we optimized our random forest model by tuning both the number of predictors per split and the total number of trees. The optimal models used 7 predictors per split, with tree counts between 200–500. Although 500 trees achieved a slightly higher accuracy (98.4%) compared to 200 trees (98.3%), we chose 200 trees for simplicity. The variable importance plots in Figure 5 below reveal eight key predictors through the Mean Decrease Accuracy criterion: “Race”, “Height”, “Age”, “FAF”, “CALC”, “NCP”, “CH2O”, “CAEC”, and “FCVC”. These findings aligned with the important predictors from our initial decision tree model.

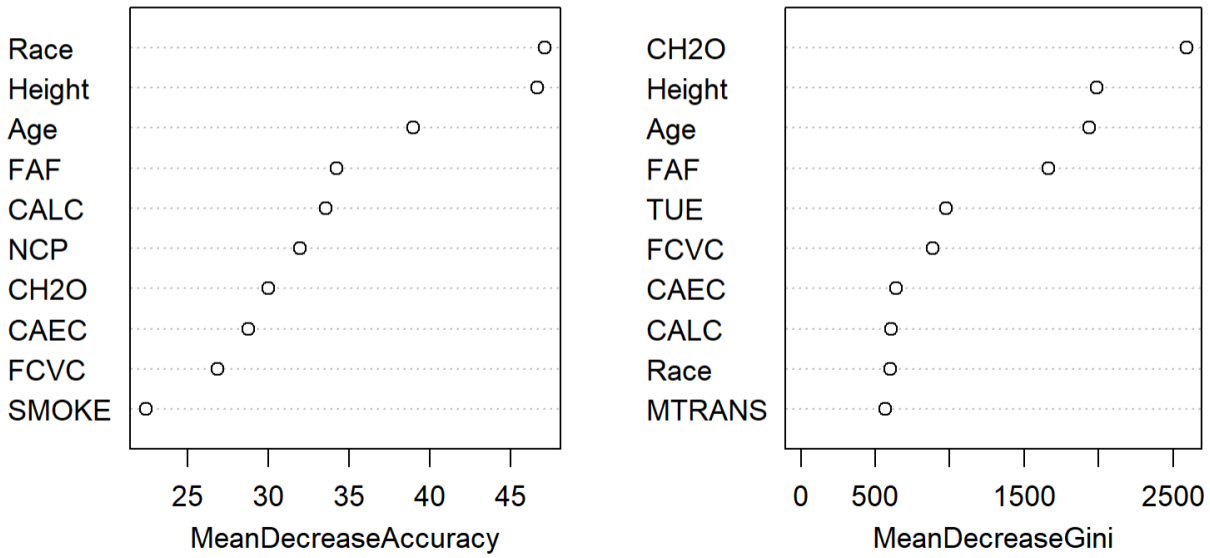

Figure 5. Importance plot for the final model (7 predictors for each tree and 200 trees)

Discussion and Limitations

Although our Random Forest model achieved a testing accuracy of 98.3%, it came with several limitations and challenges that are important to address.

While effective, random forest models operate like black boxes, making interpreting how they arrive at decisions difficult. This lack of transparency poses challenges for practical applications where the behind-the-scenes is crucial. The model's performance was also sensitive to the choice of hyperparameters and characteristics of the dataset, leading to a possible variability in results. Furthermore, since the mix of categorical and numerical predictors required extensive preprocessing, other approaches using other missing data imputation methods could have yielded different outcomes. Additionally, class imbalances in variables like stroke status and family history of overweight could have skewed predictions toward majority classes. Last but not least, our model's generalizability to other testing datasets is still uncertain, since we've only tested our model for the specific dataset. Lastly, since we can always explore more between

numerical and categorical variables in terms of interaction and other areas, we potentially leave out valuable insights.

Conclusion and Recommendations

This study shows the effectiveness of machine learning, especially Random Forest models, in predicting obesity status using a mix of categorical and numerical predictors. By integrating these variables, we achieved high accuracy, showing the importance of the integration of all factors. Random Forest consistently outperformed models like logistic regression and KNN in terms of accuracy.

Regarding future recommendations, machine learning models hold great potential for public health applications. Specifically, we can target high-risk populations by designing targeted interventions. To make models more robust, future datasets should include more diverse populations and granular lifestyle metrics, with standardized data collection protocols for consistency. Ethical considerations are also essential and could be a potential limitation. Last but not least, ensuring outputs are more interpretable will make these tools more practical.

Acknowledgments

We would like to express our sincere gratitude to Professor Akram Almohalwas for his guidance and expertise throughout the course. His clear instruction and dedication to student learning greatly enhanced our understanding of statistical concepts. We also extend our appreciation to TA Stella Huang for her invaluable assistance and patience in helping us navigate through challenging concepts and the final project.

References

Blasco, B. V., García-Jiménez, J., Bodoano, I., & Gutiérrez-Rojas, L. (2020). Obesity and Depression: Its Prevalence and Influence as a Prognostic Factor: A Systematic Review. *Psychiatry Investigation*, *17*(8), 715–724. https://doi.org/10.30773/pi.2020.0099

Chatterjee, A., Gerdes, M. W., & Martinez, S. G. (2020). Identification of Risk Factors Associated with Obesity and Overweight—A Machine Learning Overview. *Sensors*, *20*(9), 2734. https://doi.org/10.3390/s20092734

Jeffery, R. W., & Utter, J. (2003). The Changing Environment and Population Obesity in the United States. *Obesity Research*, *11*(S10). https://doi.org/10.1038/oby.2003.221

Joshu, C. E., Boehmer, T. K., Brownson, R. C., & Ewing, R. (2008). Personal, neighbourhood and urban factors associated with obesity in the United States. *Journal of Epidemiology & Community Health*, *62*(3), 202–208. https://doi.org/10.1136/jech.2006.058321

Luckhaupt, S. E., Cohen, M. A., Li, J., & Calvert, G. M. (2014). Prevalence of Obesity Among U.S. Workers and Associations with Occupational Factors. *American Journal of Preventive Medicine*, *46*(3), 237–248. https://doi.org/10.1016/j.amepre.2013.11.002

Restrepo, B. J. (2022). Obesity Prevalence Among U.S. Adults During the COVID-19 Pandemic. *American Journal of Preventive Medicine*, *63*(1), 102–106. https://doi.org/10.1016/j.amepre.2022.01.012